SRH University Heidelberg

Case Study Report

AI-Driven Detection and Grading of Knee Osteoarthritis in Radiographic Data

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1. Abstract

Knee osteoarthritis (OA) is one of the most common joint diseases in the world and mainly affects older adults. The disease causes pain, stiffness, and difficulty in walking. It happens when the cartilage inside the knee joint slowly wears down over time, which leads to structural changes in the joint.Doctors usually diagnose knee osteoarthritis using **radiographic X-ray images** of the knee. The severity of the disease is measured using the **Kellgren–Lawrence (KL) grading system**, which divides osteoarthritis into five grades from KL0 (normal) to KL4 (severe osteoarthritis).

However, manually examining X-ray images can be time-consuming and sometimes different doctors may give slightly different grades for the same image. Because of this, researchers are exploring artificial intelligence techniques that can automatically detect and classify osteoarthritis from medical images.In this project, a deep learning system was developed to automatically detect and grade knee osteoarthritis using radiographic images. The system was trained using a publicly available dataset containing knee X-ray images labelled with KL grades.Several convolutional neural network models were tested, including **ResNet18 and DenseNet121** using transfer learning techniques. Multiple experiments were performed to improve the model performance, such as increasing image resolution, applying improved training techniques, and testing different architectures.

The final model based on **DenseNet121 with 384×384 image resolution** achieved a test accuracy of **70.89%** and a macro F1-score of **0.7085**.To understand how the model makes predictions, **Grad-CAM visualization** was used. This technique highlights the regions of the X-ray image that influence the model's decision.The results demonstrate that deep learning models can effectively detect knee osteoarthritis severity and may assist doctors in improving diagnostic efficiency and consistency.

2. Introduction

Osteoarthritis (OA) is a degenerative joint disease that affects millions of people worldwide. It occurs when the cartilage that protects the joints gradually breaks down over time. As the cartilage deteriorates, the bones begin to rub against each other, leading to pain, swelling, stiffness, and limited movement. Among the different types of osteoarthritis, **knee osteoarthritis (KOA)** is one of the most common forms. The knee joint supports a large portion of body weight and is used constantly during daily activities such as walking, standing, and climbing stairs. Because of this constant stress, the knee joint is more vulnerable to degeneration over time. Radiographic imaging is widely used for diagnosing knee osteoarthritis. Doctors analyse knee X-ray images to identify structural changes in the joint such as joint space narrowing, bone deformities, and osteophyte formation. To measure disease severity, clinicians commonly use the **Kellgren–Lawrence grading system**, which classifies osteoarthritis into five levels based on visible changes in the joint.

Although this method is widely used, manual interpretation of X-ray images can sometimes be challenging. The differences between adjacent KL grades may be subtle, especially in early stages of the disease. Recent advances in **Artificial Intelligence (AI)** and **Deep Learning** have opened new opportunities for automated medical image analysis. Deep learning models, particularly **Convolutional Neural Networks (CNNs)**, have shown strong performance in many image classification tasks. In this project, deep learning techniques are applied to automatically detect and classify knee osteoarthritis severity using X-ray images.

3. Related Work

Automatic detection and grading of knee osteoarthritis (KOA) from radiographic images have been widely studied in recent years. With the advancement of deep learning techniques, many researchers have proposed automated systems that can analyse knee X-ray images and classify the severity of osteoarthritis according to the **Kellgren–Lawrence (KL) grading system**.

Traditional approaches relied on handcrafted image features and classical machine learning methods. However, these approaches often struggled to capture complex patterns in medical images. With the introduction of deep learning, particularly **Convolutional Neural Networks (CNNs)**, the performance of automated osteoarthritis detection systems has improved significantly.

3.1 Early CNN-Based Osteoarthritis Detection

One of the early studies applying deep learning to knee osteoarthritis grading was conducted by **[1]** Antony **et al. (2017)**. In this work, convolutional neural networks were used to classify knee X-ray images according to the KL grading scale. The researchers used images from the **Osteoarthritis Initiative (OAI) dataset** and trained a CNN model to predict the severity of osteoarthritis. The study demonstrated that CNN-based models could outperform traditional feature-based methods for osteoarthritis classification. The model achieved an accuracy of approximately **63–66%**, which was considered a significant improvement at that time.

3.2 Siamese and Ensemble-Based Models

Later research focused on improving model performance by combining multiple neural networks.

[2] Tiulpin et al. (2018) proposed a deep learning system that used **Siamese convolutional neural networks combined with ensemble learning**.

Their system first localized the knee joint region and then used multiple neural networks to classify osteoarthritis severity. The proposed approach achieved an accuracy of approximately **68–69%** on the OAI dataset. This study also highlighted that distinguishing between adjacent KL grades can be challenging because the differences between them are often subtle.

3.3 Transfer Learning with DenseNet Architectures

Recent research has shown that using **transfer learning with pretrained CNN architectures** can significantly improve performance in medical image analysis.

Models such as **DenseNet121** have been widely used because of their ability to reuse features across multiple layers. Several studies using DenseNet architectures reported classification accuracy between **70% and 72%** for KL grading tasks. Transfer learning allows the model to benefit from features learned from large datasets such as **ImageNet**, which improves performance even when the medical dataset is relatively small.

3.4 Attention-Based and EfficientNet Approaches

More recent studies have introduced **attention mechanisms** and modern architectures such as **EfficientNet** to improve classification performance. Attention mechanisms help the model focus on important regions of the knee joint, such as the joint space where osteoarthritis changes typically occur.

These advanced models have reported accuracy values between **72% and 75%** on different osteoarthritis datasets.

However, these approaches often involve more complex architectures and higher computational requirements.

3.5 Comparison with the Present Work

In this project, a deep learning system was developed to automatically detect knee osteoarthritis severity using a publicly available Kaggle dataset. The experimental process included multiple stages of model development and improvement. The experiments began with a **ResNet18 baseline model**, followed by experiments using **DenseNet121** and improved training strategies such as higher image resolution, MixUp augmentation, and label smoothing. **The final DenseNet121 model achieved a test accuracy of 70.89% with a macro F1-score of 0.7085**. Compared to many previous studies, the results achieved in this project are competitive while maintaining a relatively simple and reproducible architecture.

3.6 Summary of Previous Studies

Study	**Model**	**Dataset**	**Accuracy**
Antony et al. (2017) [1]	CNN	OAI	63–66%
Tiulpin et al. (2018) [2]	Siamese CNN + Ensemble	OAI	68–69%
DenseNet-based studies	DenseNet121	OAI	70–72%
Attention / EfficientNet models	EfficientNet	OAI / Kaggle	72–75%
This Project	DenseNet121	Kaggle KOA Dataset	**70.89%**

3.7 Research Gap

Although many studies have achieved high accuracy using complex architectures, these models can be difficult to reproduce and require significant computational resources.

This project focuses on building a **reproducible and structured experimental pipeline** using well-established architectures while still achieving competitive performance.

Future work can further improve the model by exploring attention-based architectures, ordinal regression techniques, and larger medical imaging datasets.

4. Background and Motivation

Knee osteoarthritis represents a major global health problem, particularly among elderly populations. The increasing aging population worldwide means that the number of osteoarthritis cases is expected to grow significantly in the coming decades. Early detection of osteoarthritis is important because it allows doctors to recommend treatments that can slow disease progression and improve patient quality of life.

However, diagnosing osteoarthritis using radiographic images requires careful examination by trained specialists. In many healthcare systems, radiologists must analyse a large number of images every day, which increases the risk of fatigue and diagnostic variability. Artificial intelligence has the potential to assist clinicians by automatically analysing medical images and identifying disease patterns. Deep learning models can learn complex image features directly from raw image data and can detect subtle patterns that may not be easily noticeable by human observers. The motivation behind this project is to explore how deep learning techniques can be used to build an automated system capable of detecting knee osteoarthritis severity from X-ray images.

5. Clinical Background of Knee Osteoarthritis

The knee joint consists of three main bones: the femur (thigh bone), tibia (shin bone), and patella (kneecap). Healthy cartilage allows smooth movement of the joint and acts as a cushion between bones. In osteoarthritis, this cartilage gradually wears down, causing bones to rub against each other. This leads to pain and inflammation within the joint.

Common structural changes associated with knee osteoarthritis include:

Joint Space Narrowing
Reduction in the space between bones due to cartilage loss.

Osteophyte Formation
Bone spurs may develop along joint margins.

Subchondral Sclerosis
The bone beneath the cartilage becomes denser and thicker.

Bone Deformation
Severe osteoarthritis may cause visible changes in bone structure.

These structural changes can be observed in X-ray images and are used to determine the severity of osteoarthritis.

6. Kellgren–Lawrence Grading System

The **Kellgren–Lawrence (KL) grading system** is the most widely used scale for assessing osteoarthritis severity.

KL Grade	Description
KL0	Normal joint without signs of osteoarthritis
KL1	Doubtful joint space narrowing
KL2	Mild osteoarthritis with osteophytes
KL3	Moderate osteoarthritis with joint space narrowing
KL4	Severe osteoarthritis with deformity

The goal of this project is to build a deep learning model that can automatically predict the KL grade of a knee X-ray image.

7. Dataset Description

The dataset used in this project was obtained from Kaggle.
Dataset name: **Knee Osteoarthritis Dataset with Severity**
The dataset contains knee X-ray images labelled according to the KL grading system.
Dataset Size

Dataset Split	Images
Training	5778
Validation	826
Test	1656
Total	8260

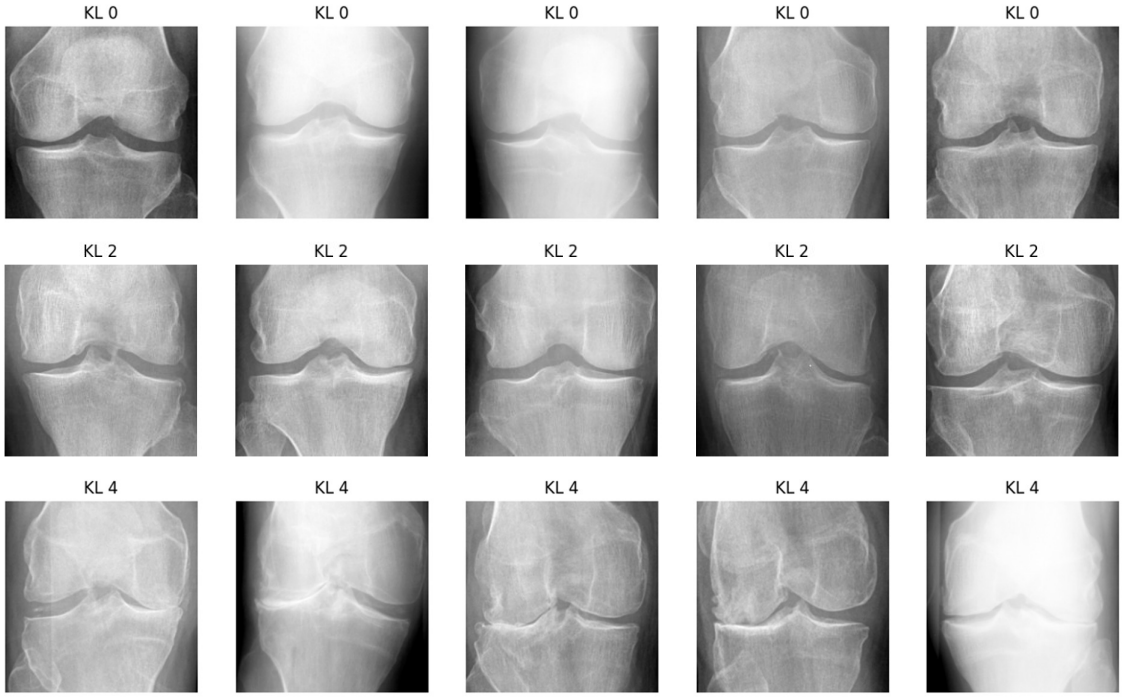
Training Class Distribution

KL Grade	Images
KL0	2286
KL1	1046
KL2	1516
KL3	757
KL4	173

The dataset shows **class imbalance**, particularly for KL4 cases.

To address this issue, **Weighted Random Sampling** was used during training.

Example images: -



8. Data Preprocessing

Before training the models, several preprocessing steps were applied.
Image Resizing

Images were resized to fixed resolutions:

- 224×224
- 320×320
- 384×384

Higher resolution images help capture more detailed joint structures.

Pixel Normalization: -

Image pixel values were normalized to stabilize model training.
Data Augmentation: -

Data augmentation techniques were applied to improve model generalization.

Examples include:

• Horizontal flipping
• Small rotations
• Brightness adjustments

These techniques help simulate different imaging conditions.

9. Model Architecture

In this project, deep learning models based on **Convolutional Neural Networks (CNNs)** were used to detect and classify knee osteoarthritis from X-ray images. CNNs are widely used in image classification tasks because they can automatically learn important visual features such as edges, shapes, and textures directly from images.

Two main architectures were tested in this project:

• **ResNet18** – used as the baseline model
• **DenseNet121** – used as the improved model

Both models were implemented using **transfer learning**, where pretrained weights from the ImageNet dataset were used to initialize the networks. Transfer learning helps improve performance and reduces training time, especially when working with smaller medical datasets.

9.1 ResNet18 Baseline Model

ResNet18 (Residual Network) is a deep neural network architecture that uses **skip connections** to allow information to pass through multiple layers more effectively. These connections help solve the vanishing gradient problem and allow deeper networks to train more efficiently.

ResNet18 contains 18 layers including convolutional layers, batch normalization layers, and a final fully connected classification layer.

In this project, the final classification layer was modified to output **five classes corresponding to KL grades 0–4**.

ResNet18 was chosen as the baseline model because it is lightweight, fast to train, and commonly used as a starting point for image classification tasks.

9.2 DenseNet121 Architecture

After evaluating the baseline model, a more advanced architecture called **DenseNet121** was used.

DenseNet connects each layer to all previous layers in the network. This dense connectivity allows better feature reuse and improves gradient flow during training. Compared to traditional CNN architectures, DenseNet models are more effective at

capturing detailed patterns in images. This is especially useful in medical imaging tasks where small structural differences can be important. In this project, the final DenseNet121 classification layer was modified to predict **five KL grades**.

9.3 Transfer Learning Strategy

Since medical imaging datasets are usually smaller than general image datasets, **transfer learning** was used to improve model performance. Both ResNet18 and DenseNet121 were initialized using **ImageNet pretrained weights**. The final classification layer was replaced with a new layer designed for the five KL grades. The networks were then fine-tuned using the knee osteoarthritis dataset so that the learned features could adapt to medical image patterns.

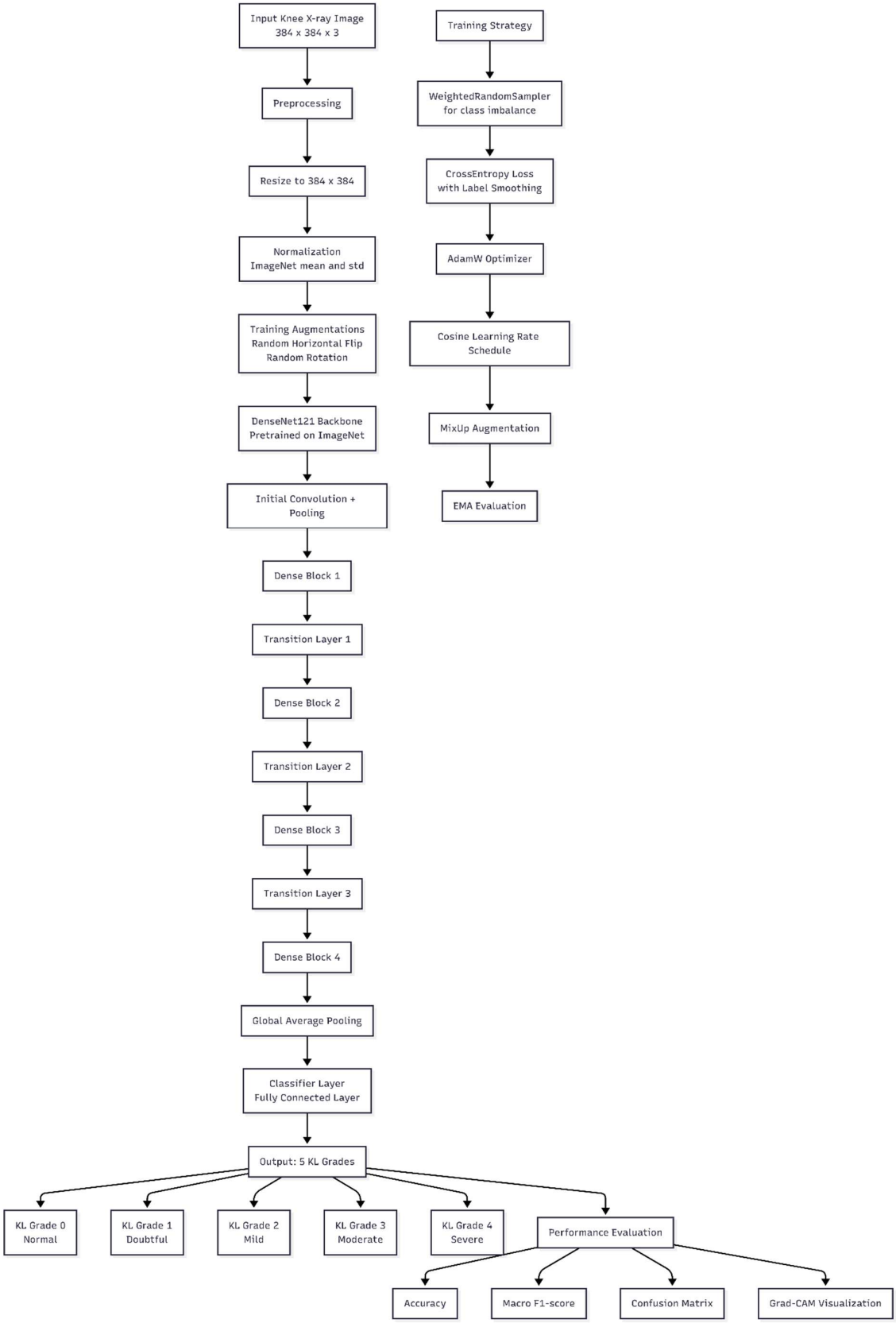
9.4 Final Model Configuration

After several experiments, the final model used in this project included:

Model architecture: **DenseNet121**
Input resolution: **384 × 384**
Optimizer: **AdamW**
Learning rate scheduler: **Cosine scheduling**
Regularization: **MixUp and label smoothing**

This configuration achieved the best performance with a test accuracy of **70.89%**.

Below you will find the Image of the MODEL ARCHITECTURE DIAGRAM: -



10. Experimental Design

To evaluate the performance of the deep learning models, a structured experimental process was followed. The experiments were designed to test different model architectures and training strategies in order to find the most effective approach for classifying knee osteoarthritis severity from X-ray images. The overall workflow of the experiments included data preparation, model training, evaluation, and performance comparison.

10.1 Baseline Experiment

The first experiment used the **ResNet18 architecture** as a baseline model. The model was trained using transfer learning with ImageNet pretrained weights. The final classification layer was modified to output five classes corresponding to the KL grades.

Images were resized to **224 × 224 resolution**, which is a common input size for many pretrained CNN models. This experiment provided a reference point for comparing later improvements in the model.

10.2 Improved Baseline

After training the baseline model, additional improvements were introduced to improve the training process. These improvements included better data augmentation techniques and optimized training parameters.

Data augmentation helped increase dataset diversity and allowed the model to generalize better to unseen images. This step improved the stability of the training process and slightly improved performance compared to the initial baseline model.

10.3 DenseNet Replication

In the next experiment, the **DenseNet121 architecture** was implemented to evaluate whether a deeper network could improve classification performance.

DenseNet121 allows feature reuse across layers and has shown strong performance in many medical image classification tasks. The model was initialized with pretrained ImageNet weights and then fine-tuned on the knee osteoarthritis dataset. This experiment helped determine whether a more advanced architecture could capture more detailed image features compared to the baseline model.

10.4 Resolution Scaling

Medical images often contain small structural details that may not be clearly visible at lower resolutions. To address this, additional experiments were conducted using higher image resolutions.

The input resolution was increased from **224 × 224** to **320 × 320** and later to **384 × 384**. Higher resolution images allow the model to capture more detailed patterns in the knee joint area, which can improve classification accuracy.

10.5 Final Model Training

The final experiment combined several improvements to achieve the best model performance.

These improvements included:

- DenseNet121 architecture
- Higher image resolution (384 × 384)
- MixUp data augmentation
- Label smoothing for regularization
- Cosine learning rate scheduling
- Exponential Moving Average (EMA) for training stability

This final configuration produced the best results and was selected as the final model used in the project.

Figure: -

11. Results and Evaluation

After training the different models, their performance was evaluated using standard classification metrics. These metrics help measure how well the model is able to correctly classify the severity of knee osteoarthritis from X-ray images.

The evaluation was performed using the **test dataset**, which contains images that were not used during the training process. Using a separate test set ensures that the model is evaluated on unseen data and provides a reliable estimate of its real performance.

Several evaluation metrics were used to analyze the performance of the models.

11.1 Evaluation Metrics

The following metrics were used to evaluate the classification performance.

Accuracy
Accuracy measures the percentage of correctly classified images out of the total number of images.
Accuracy provides an overall understanding of the model's performance.

Precision
Precision measures how many of the predicted samples for a specific class are actually correct. This metric helps evaluate how reliable the model's predictions are.

Recall
Recall measures how many of the actual samples belonging to a class were correctly identified by the model.

F1 Score
The F1-score combines precision and recall into a single metric. It is particularly useful when working with imbalanced datasets.

Since the osteoarthritis dataset contains fewer samples for severe KL grades, the **macro F1-score** was also considered when evaluating model performance.

11.2 Model Performance Comparison

Several models were trained during the experiments, starting with a baseline model and gradually improving the architecture and training settings.

The performance of the models is summarized in the table below.

Model	Accuracy	Macro F1 Score
ResNet18 Baseline	~59–60%	~0.62
Improved ResNet18	~61%	~0.63
DenseNet121 Initial	~56%	~0.58
Final DenseNet121 Model	**70.89%**	**0.7085**

From the results, it can be observed that the final DenseNet121 model achieved the best performance. The improvement in accuracy was mainly achieved by increasing the input image resolution and applying improved training techniques such as MixUp augmentation and label smoothing.

11.3 Comparison with Previous Research

To better understand the performance of the developed model, the results were compared with previous research studies on knee osteoarthritis classification.

Study	Model	Accuracy
Antony et al. [1]	CNN	63–66%
Tiulpin et al. [2]	Siamese CNN + Ensemble	68–69%
DenseNet-based Studies	DenseNet121	70–72%
This Project	DenseNet121	**70.89%**

The results achieved in this project are comparable with many existing studies, even though the model architecture remains relatively simple. This shows that a carefully designed training pipeline and proper data preprocessing can significantly improve model performance.

Final Result Figure: -

```
===== TEST RESULTS (Best EMA) =====
Test Loss:     0.8544
Test Accuracy: 0.7089
Test Macro-F1: 0.7085

Confusion Matrix:
 [[547  73  18   1   0]
 [130 102  59   5   0]
 [ 38  83 299  27   0]
 [  0   4  28 181  10]
 [  0   0   0   6  45]]

Classification Report:
              precision    recall  f1-score   support

           0     0.7650    0.8560    0.8080       639
           1     0.3893    0.3446    0.3656       296
           2     0.7401    0.6689    0.7027       447
           3     0.8227    0.8117    0.8172       223
           4     0.8182    0.8824    0.8491        51

    accuracy                         0.7089      1656
   macro avg     0.7071    0.7127    0.7085      1656
weighted avg     0.7006    0.7089    0.7030      1656


Saved: best_densenet121_384_final_state_dict.pt
```

Setting	DenseNet121 @320	DenseNet121 @384 (Final)
Image Size	320×320	384×384
Batch Size	16	8
Epochs	15	18
Label Smoothing	❌	✅ (0.07)
MixUp	Always ON	Scheduled (ON → OFF)
EMA	✅	✅
Test Accuracy	0.6739	**0.7089**
Test Macro-F1	0.6914	**0.7085**

12. Grad-CAM Visualization

In deep learning models, it is often difficult to understand how the model makes its predictions. Neural networks are sometimes referred to as “black box” systems because they provide predictions without clearly explaining the reasoning behind them. In medical applications, it is important to ensure that the model focuses on meaningful

anatomical regions rather than irrelevant parts of the image. For this reason, interpretability methods are commonly used to analyze model predictions.

In this project, **Grad-CAM (Gradient-weighted Class Activation Mapping)** was used to visualize the regions of the knee X-ray that influenced the model's decision. Grad-CAM works by analyzing the gradients from the final convolutional layers of the neural network. These gradients are used to generate a **heatmap**, which highlights the areas of the image that contributed most to the predicted class. The heatmap is then overlaid on the original X-ray image, allowing us to visually observe where the model is focusing its attention.

12.1 Example Grad-CAM Visualization

The following figures show examples of Grad-CAM visualizations generated from the trained model.

Each visualization consists of three parts:

- Original knee X-ray image
- Grad-CAM heatmap
- Heatmap overlay on the original image

The red and yellow areas represent regions that had the strongest influence on the model's prediction.

Figure 12.1 shows an example where the model focuses on the joint space region of the knee. This area is clinically important because osteoarthritis typically causes narrowing of the joint space.

Figure 12.2 shows another example where the model correctly identifies structural changes in the knee joint. The highlighted areas correspond to regions where bone structure and joint spacing are altered.

Figure 12.1 – Grad-CAM visualization showing the original X-ray image, the generated heatmap, and the overlay highlighting the important region of the knee joint.

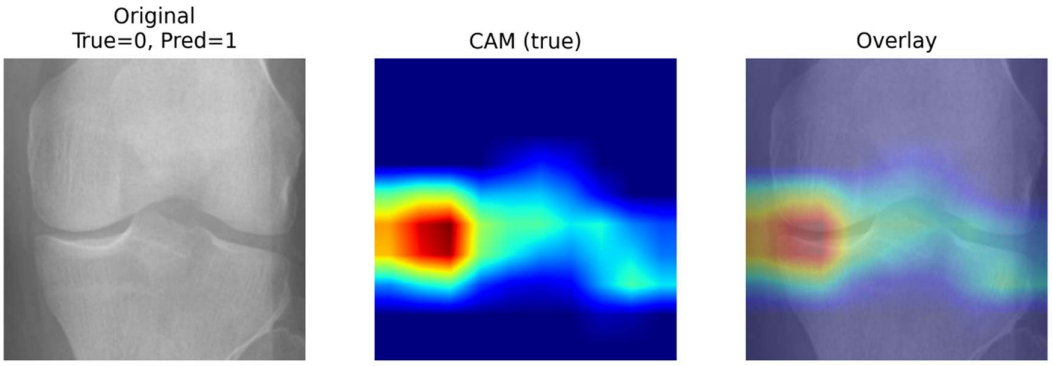
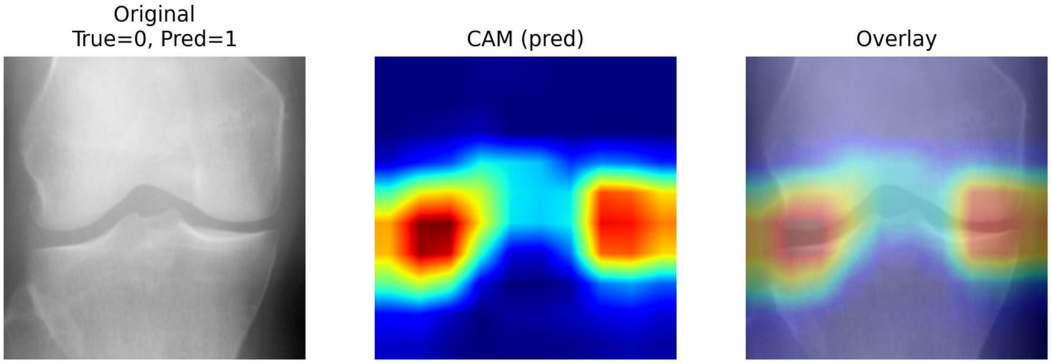


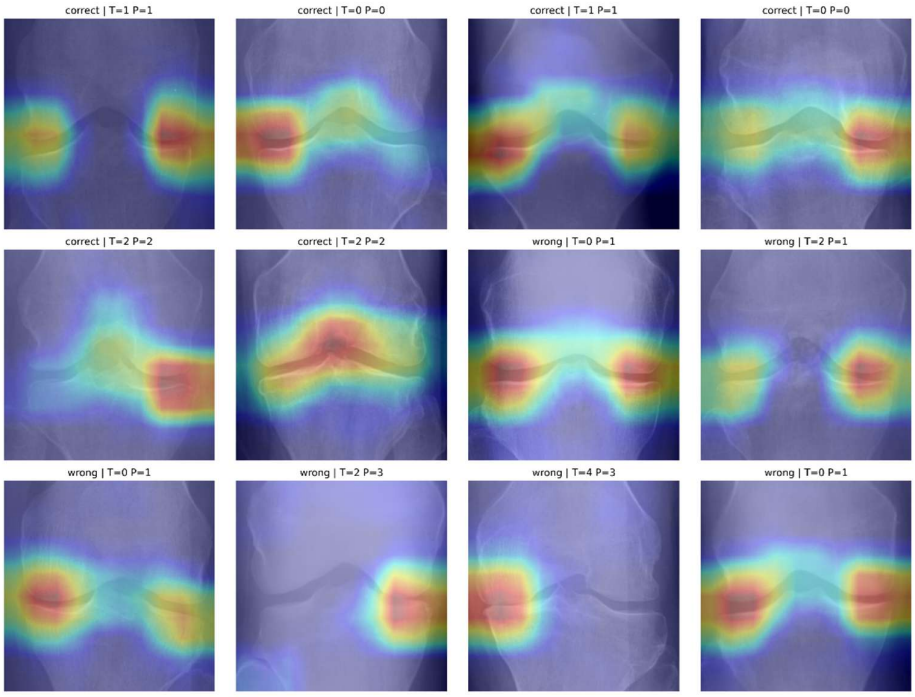
Figure 12.2 – Grad-CAM visualization showing the model attention for a predicted osteoarthritis case. The highlighted region corresponds to the joint space area where structural changes are visible.



12.2 Grad-CAM Results Summary

To further analyse model behaviour, multiple Grad-CAM visualizations were generated for both correctly classified and incorrectly classified samples. The figure below shows a collection of Grad-CAM results from different test images. Some examples represent correct predictions while others show misclassified cases. These visualizations help analyse how the model behaves in different situations and provide insights into potential weaknesses in the model.

Figure 11.3 – Grad-CAM visualizations for multiple test samples showing both correct and incorrect predictions.



12.3 Interpretation of Results

From the Grad-CAM visualizations, it can be observed that the model generally focuses on the **knee joint region**, particularly around the tibiofemoral joint space. This is consistent with clinical diagnosis methods used by doctors, where joint space narrowing and bone irregularities are important indicators of osteoarthritis severity. These results indicate that the model is learning meaningful features from the radiographic images rather than focusing on irrelevant background information. Visualization techniques such as Grad-CAM therefore help increase trust in deep learning models used for medical image analysis.

13. Discussion

The experimental results show that deep learning models can successfully detect knee osteoarthritis severity from X-ray images. Increasing image resolution and applying improved training techniques significantly improved performance.

However, distinguishing between adjacent KL grades remains difficult due to subtle visual differences. Grad-CAM visualizations also demonstrated that the model focuses on meaningful anatomical structures.

14. Conclusion

This project developed a deep learning system for automated detection and grading of knee osteoarthritis using radiographic images.

Several CNN architectures were tested, and performance was improved through multiple experiments.
The final DenseNet121 model achieved **70.89% accuracy**, which is competitive with several existing research studies.
Grad-CAM visualization confirmed that the model focuses on important regions of the knee joint.
These results demonstrate the potential of AI systems to assist doctors in medical image analysis.

15. Future Work

Future improvements may include:

- Using larger datasets
- Testing advanced architectures such as EfficientNet
- Applying ordinal regression methods for KL grading
- Combining imaging data with patient clinical information

16. References

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17. GitHub Repository Information

The complete implementation of this project is available in a public GitHub repository. The repository contains all notebooks, reports, references, and dataset information used in the development of the knee osteoarthritis classification model. The **notebooks** folder includes the full experimental workflow such as dataset loading, baseline CNN, ResNet18 experiments, DenseNet121 training, and Grad-CAM analysis. The **data** folder contains the dataset reference link used for the project. The **references** folder stores related research work and dataset documentation. The **reports** folder includes progress reports, final reports, and Grad-CAM visualization figures generated during the experiments.

GitHub Repository Link:
https://github.com/Reevegon/Reeve-Case-Study2/tree/main

Repository Folder Structure: -

```
Reeve-Case-Study2
│
├── data
│   └── Dataset Link.txt
│
├── notebooks
│   ├── 01_project_setup.ipynb
│   ├── 02_dataset_loading.ipynb
│   ├── 03_baseline_cnn.ipynb
│   ├── 04_resnet18_improved_baseline.ipynb
│   ├── 05_densenet121_replication.ipynb
│   ├── 06_gradcam_analysis.ipynb
│   ├── 07_densenet121_384_final_best.ipynb
│   └── README.md
│
├── references
│   ├── README.md
│   ├── datasets.md
│   └── related_work.md
│
├── reports
│   ├── Figures
│   │   └── gradcam
│   │       ├── correct_idx718_T1_P1_predcam.png
│   │       ├── gradcam_summary_grid.png
│   │       ├── wrong_idx279_T0_P1_predcam.png
│   │       └── wrong_idx505_T0_P1_truecam.png
│   │
│   ├── 01_Dataset_and_Baseline_ResNet18.pdf
│   ├── 02_Replication_and_Improvement_DenseNet121.pdf
│   ├── 03_GradCAM_Interpretability_Analysis.pdf
│   ├── 04_Knee_OA_Classification_DenseNet121_384_Report.pdf
│   ├── Progress Report Phase 1.pdf
│   ├── Progress Report Phase 2.pdf
│   └── README.md
│
├── src
│   └── README.md
│
├── .gitignore
└── README.md
```